

Along-Slit Stress Test: Pipeline & Assumptions

A conceptual walkthrough of how the synthetic scenes, the detector projection, and the three retrieval strategies fit together — and what four bands of realistic along-slit truth reveal about geometric-distortion bias.

4 spectral bands 3 synthetic scenes 3 retrieval strategies

1024 detector rows per band

1 Why distort a synthetic scene at all?

Every GeoCarb detector carries two well-characterized geometric imperfections in its optics.

Keystone: the true along-slit position a given pixel is really looking at drifts across the columns of a single detector row, instead of staying fixed for the whole row. **Smile:** the wavelength each column corresponds to bends slightly from row to row, instead of following one clean dispersion law for the whole array. Both are real, both are measured on the ground, and both are small — at most a few pixels' worth of drift.

The question this study asks is simple to state and hard to answer from first principles: if the true atmosphere and surface actually vary along the slit, how much of that variation leaks into the retrieved gas amounts purely because of these two distortions — before any real atmospheric signal is even in play? Answering it means building a synthetic scene whose truth is known exactly, pushing it through a detector model that reproduces the real distortions pixel for pixel, and retrieving it back with three different levels of geometric awareness.

2 Three scenes, three assumptions

All three scenes share the same detector, the same four bands, and the same retrieval machinery. What changes is only the truth painted onto the slit before it ever reaches the optics.

UNIFORM

A flat truth atmosphere at every along-slit position — no real composition or pressure signal anywhere. Isolates pure geometric-distortion bias, with nothing else around to explain a nonzero result.

BARCODE

Still a flat composition truth, but illumination brightness alternates in sharp bars along the slit. Nothing about the atmosphere changes — only brightness. Built to expose keystone/point-spread mixing right at a sharp transition, where a real scene boundary (a cloud edge, a coastline) would sit.

REALISTIC

The full along-slit truth: continuously varying CO₂, CH₄, CO, H₂O, and surface pressure, including a few sharp, narrow features layered on smoother large-scale gradients. The case meant to look like an actual overpass — and the one behind every figure in §7.

3 Building the detector signal: native

The starting point for every scene is a spectrum computed at full spectral resolution as a function of true along-slit position only — no detector, no pixels, no distortion yet. Call this the **object-space spectrum**: for any true position on the slit, it says exactly what radiance reaches the front of the instrument at every wavelength.

Turning that into what a real detector pixel records takes one operator, applied pixel by pixel: the **projection operator**. It is the real, ground-measured mapping that says exactly which (wavelength, true along-slit position) pair a given physical pixel is looking at — the same mapping that encodes both keystone and smile, since it was fit directly to laboratory calibration data rather than to an idealized optical design. For a single detector row, this is not one clean wavelength axis over one

fixed slit position: because of keystone, different columns of the same row can correspond to noticeably different true along-slit positions, especially near the top and bottom of the slit.

Once the projection operator has picked out each pixel's own (wavelength, true position) pair, the instrument's spectral response — its line shape, or **ILS** — is convolved directly at that pixel's own wavelength, against the object-space spectrum evaluated at that pixel's own true position.

Projection and instrument-resolution blurring happen together, in one pass per pixel, not as two separate stages: there is no shared, already-resampled spectrum for a whole row to fall back on, because different columns of a keystone-affected row are genuinely looking at different places on the slit.

The one remaining step is a real cross-row blur along the north–south, along-slit direction — the instrument's actual spatial point-spread function — the only place where neighboring detector rows are allowed to mix. Everything upstream of that blur is evaluated pixel by pixel, with no borrowing between rows.

Native retrieves directly from this fully-distorted detector row, exactly as built, using each pixel's own real wavelength as the retrieval's wavelength grid. It sees the same keystone, smile, and spatial blur a real instrument would, with no attempt to remove or correct for any of it beforehand.

4 Rectification: resampling onto a shared grid

A real ground-processing pipeline does not retrieve directly off the raw distorted image the way native does. Instead, it **rectifies** the raw detector data first: it defines one shared, regular grid of (along-slit position, wavenumber) values common to the whole array, and for every point on that regular grid, it asks the projection operator's *inverse* where that combination of position and wavenumber actually landed on the real, distorted detector — then interpolates the raw pixel values found there. The result is a tidy, regular spectrogram that looks like it came from a perfectly well-behaved instrument, built entirely after the fact from data that never had that property to begin with.

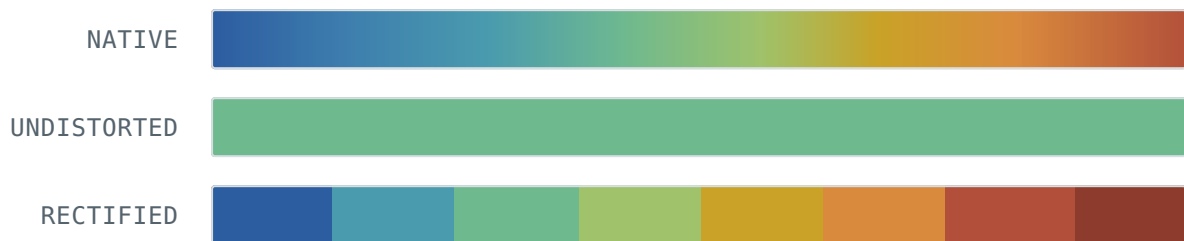
This step is not free. Because keystone and smile mean the real mapping and its inverse are not perfectly smooth, single-valued functions across the whole array, interpolating onto a shared grid necessarily blends raw pixels that were really looking at slightly different positions and wavelengths. That blending is a genuine, separate source of bias — on top of whatever geometric distortion the

rectified pipeline was trying to remove in the first place. **Rectified** is exactly native's row, resampled this way, then retrieved on the shared grid instead of each row's own real one.

5 Undistorted: the zero-keystone counterfactual

The third strategy asks a narrower question: how much of native's bias is really about keystone specifically, versus something more fundamental about the band itself? **Undistorted** keeps the real per-row wavelength scale — including the real smile curve, i.e. the genuine row-to-row bending of the wavelength-to-column relationship measured on the ground — but evaluates every pixel in a row at that row's own single, fixed along-slit position, rather than letting each column sample a slightly different true position the way keystone genuinely does.

In effect: the same detector, with the same real dispersion law and the same row-to-row smile shift baked in, just squared off along the slit direction so a row is a clean single position instead of a diagonal cut across the true scene. There is no resampling step here and no spatial point-spread blur either — undistorted bypasses the projection/rectification machinery entirely and evaluates the object-space spectrum directly, at each row's fixed position, on that row's own real wavelength axis. Any bias that still shows up in undistorted is not a keystone artifact — it has to be a genuine limit of what this band's spectrum can tell you, present even with a perfectly well-behaved detector.



Schematic, not to scale: one detector row's true along-slit position, column by column, under each strategy. Color encodes true position. Native sweeps continuously across whatever positions keystone really puts under that row; undistorted is one clean position for the whole row; rectified is native's sweep cut into discrete resampled blocks, with a seam at every block boundary.

6 The retrieval: single-band, and what's in the state vector

Every retrieval in this study is **single-band**: each of GeoCarb's four bands — the O₂ A-band, the weak CO₂ band, the strong CO₂ band, and the combined CH₄/CO band — is retrieved completely independently, row by row, with no information shared across bands.

For every row, the retrieval fits the same family of state-vector elements regardless of band:

ELEMENT	APPLIES TO	WHAT IT DOES
Target gas scale factor(s)	every band	Multiplies the prior gas column to match the row's data — CO ₂ for the two CO ₂ bands, CH ₄ and CO together for the combined band.
Water-vapor scale factor	every band	Retrieved everywhere, since water vapor absorbs across the whole spectral range GeoCarb covers, not just in one dedicated band.

ELEMENT	APPLIES TO	WHAT IT DOES
Surface-pressure scale factor	every band	The one term every band actually senses total air mass through — for the O ₂ band in particular, this is the whole story (see note below).
Temperature offset	every band	A single scalar shift applied to the whole assumed temperature profile.
Surface albedo, level + slope	every band	Surface brightness and its mild spectral slope across the band.
Low-order dispersion correction	native, rectified	A small polynomial nudge to the assumed wavelength calibration, standing in for the residual calibration uncertainty a real instrument always carries. Left out of undistorted, which has no genuine calibration error for it to absorb.

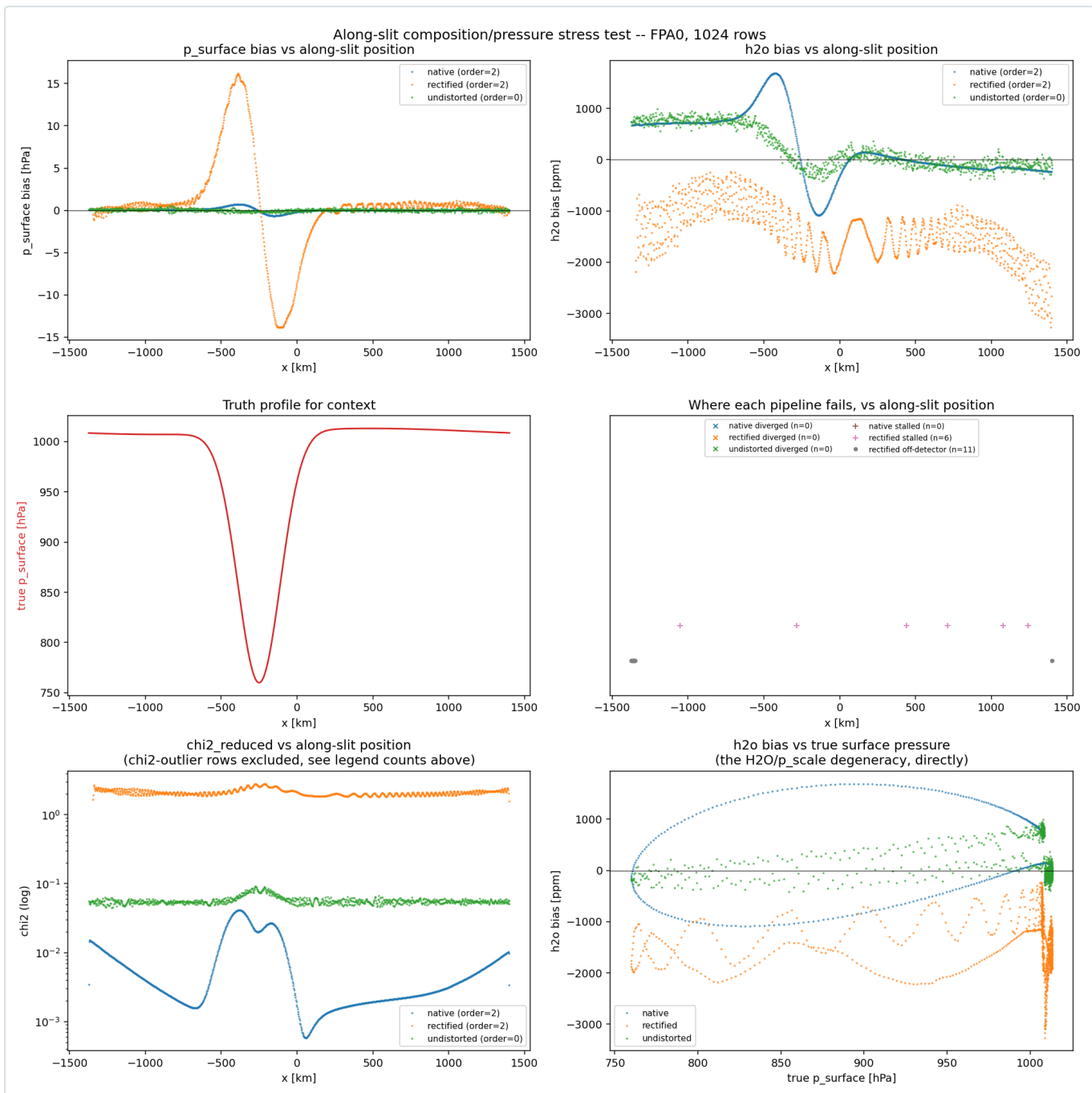
Well-mixed gases like O₂ are deliberately excluded from getting their own scale factor: because their true column doesn't vary along the slit at all, floating them separately would be nearly indistinguishable from floating surface pressure, and the fit has nothing to break that tie with. Pressure alone carries that signal instead — exactly as an operational O₂ A-band retrieval does.

Every row is also retrieved against the **same fixed prior atmosphere** — the true state at the center of the slit — never each row's own true state. Whatever a row's truth deviates from that fixed center is exactly what its gas-scale and pressure-scale terms have to recover.

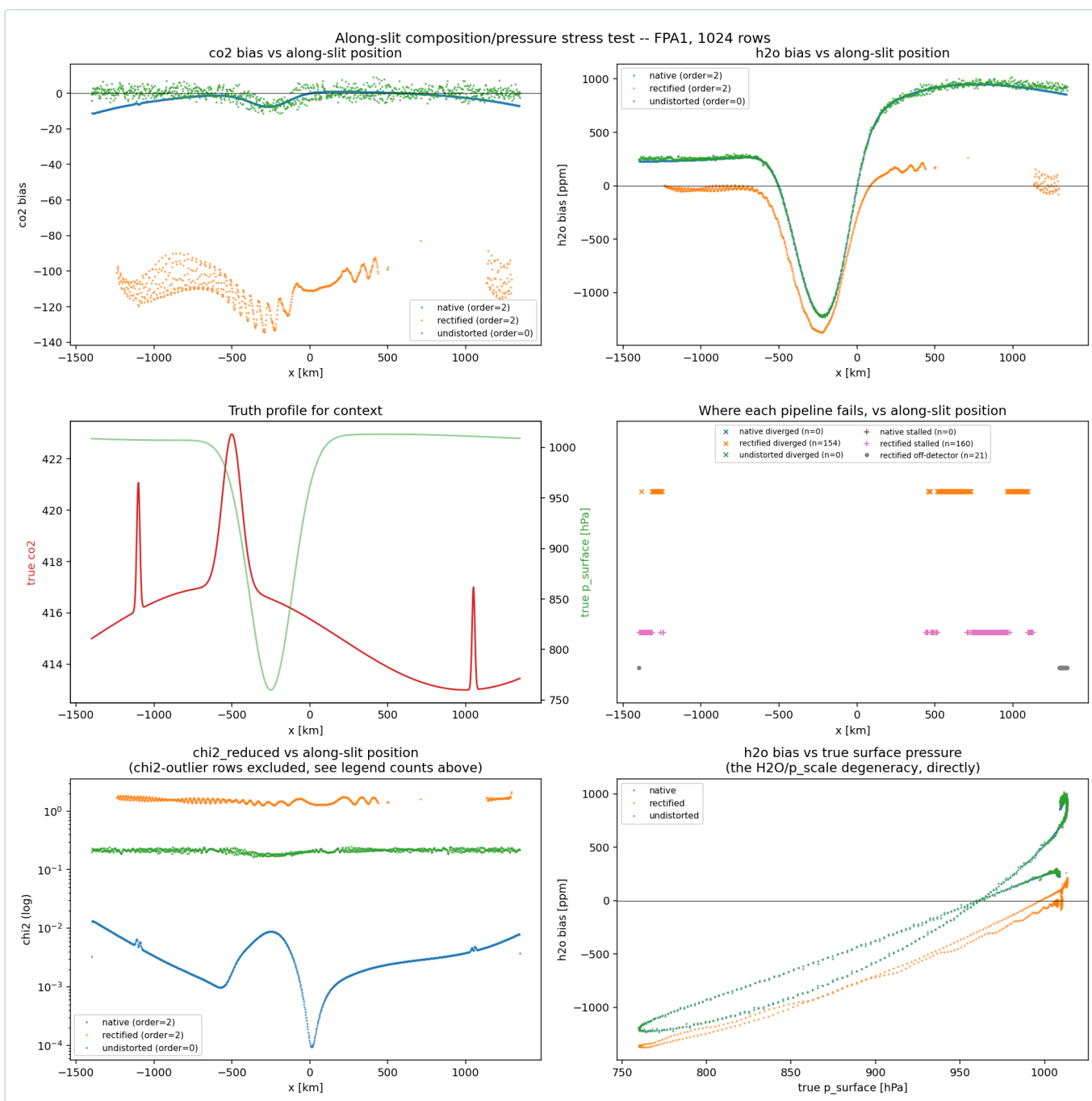
7 Results: bias vs. along-slit position, realistic scene

The four figures below are the realistic scene's summary diagnostic for each band: retrieved bias (retrieved minus true) for every one of 1024 detector rows, plotted against real along-slit position, for all three strategies at once — the top two panels in each figure. The remaining panels in each

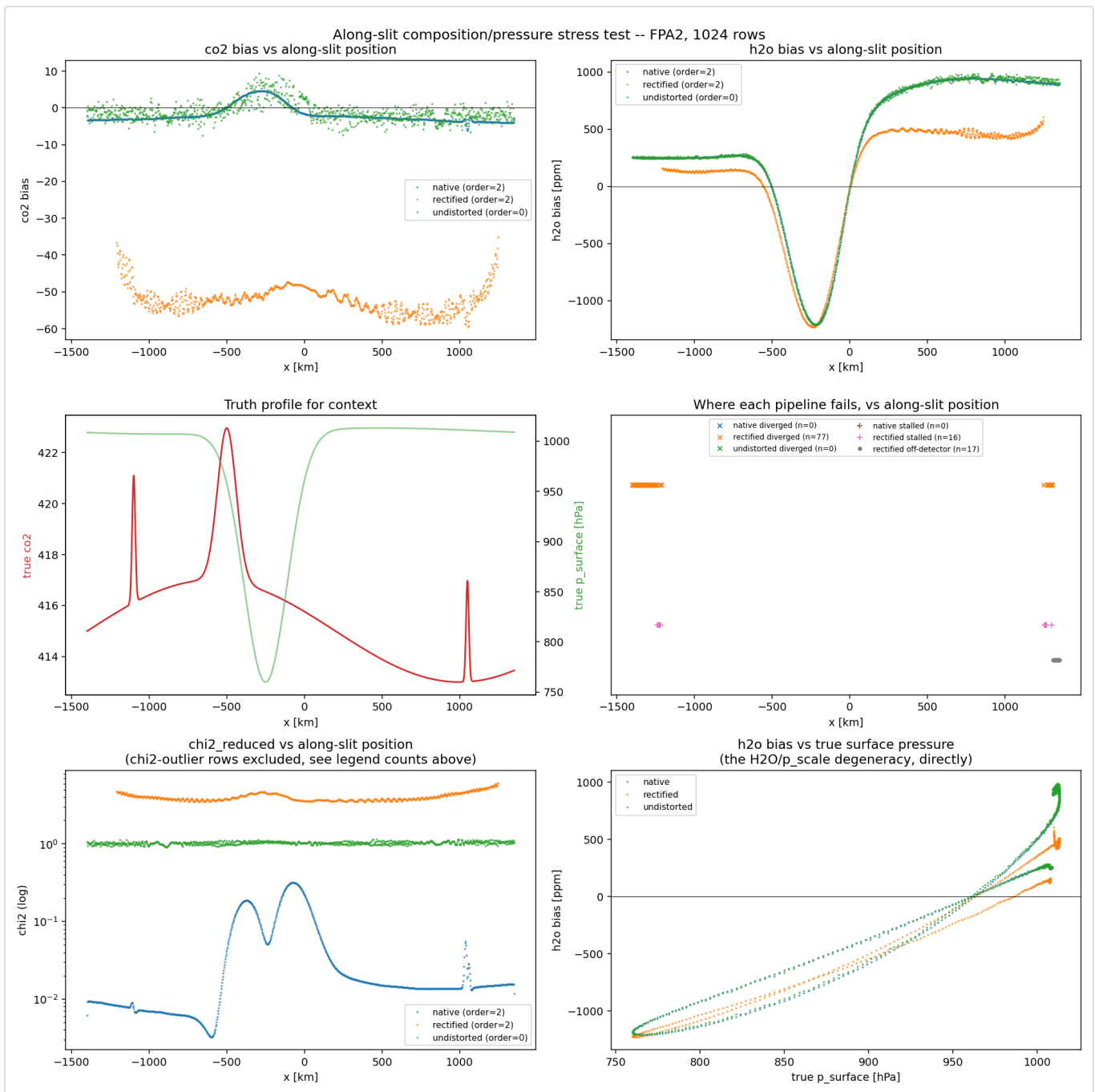
figure carry supporting context: the truth profile behind the row, where each pipeline fails along the slit, reduced chi-squared, and the water-vapor/surface-pressure degeneracy plotted directly against true pressure.



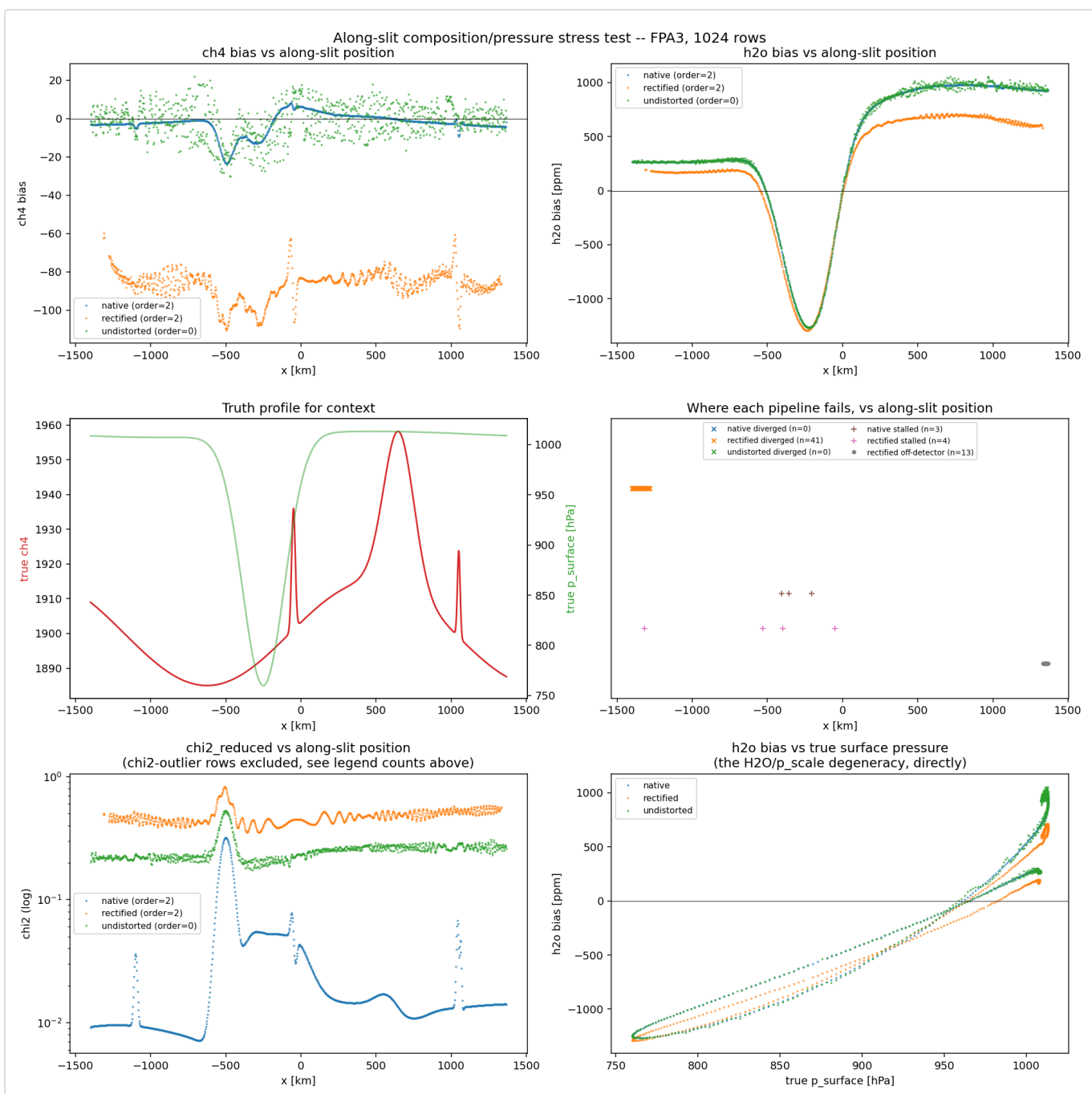
FPA0 · O₂ A-band — retrieves H₂O + surface pressure only (O₂ itself excluded, see §6). Native and undistorted track the true surface-pressure feature to within about a hPa; rectified overshoots by up to 15 hPa at the same feature and drifts further at both slit extremes.



FPA1 · weak CO₂ band — retrieves CO₂ + H₂O. Native and undistorted stay within roughly ± 10 ppm of true CO₂ almost everywhere; rectified sits at a roughly constant -100 ppm offset across the whole slit, on top of a real dip all three pipelines share near the pressure feature.



FPA2 · strong CO₂ band — retrieves CO₂ + H₂O. Same qualitative split as the weak band, at a smaller absolute offset (rectified around -40 to -60 ppm) — this band's stronger absorption gives the fit more to work with.



FPA3 · combined CH₄/CO band — retrieves CH₄, CO, H₂O. Native and undistorted again track truth closely except at one real, shared feature; rectified's CH₄ bias sits offset by roughly -60 to -100 ppb across the slit.

COMMON THREAD

In every band, native and undistorted trace almost exactly on top of each other, including through one real shared feature — a water-vapor/surface-pressure degeneracy, visible in the lower-right panel of each figure — that persists even with zero detector distortion, meaning it's a genuine limit of the band's information content, not an artifact. Rectified's bias sits apart from both, at a scale set mostly by the resampling step itself, not by keystone or smile directly.

8 A closer look: why does native's fit quality look so much better?

One number in the figures above is easy to misread: native's reduced chi-squared runs one to two orders of magnitude below undistorted's, at nearly every row. Taken at face value that looks backwards — undistorted is the zero-distortion baseline, so it might be expected to fit at least as well as native, not dramatically worse.

Testing this directly turns out to be simple: retrieve the same rows with native's and undistorted's roles swapped — give undistorted the same wavelength-dispersion correction native normally gets, and take it away from native. Matched at the same freedom, the gap nearly vanishes:

ROW	NATIVE, ITS USUAL SETTING	UNDISTORTED, ITS USUAL SETTING	NATIVE, WITHOUT THE CORRECTION	UNDISTORTED, WITH THE CORRECTION
0	0.0028	0.163	0.653	0.0010
512 (slit centre)	0.0003	0.161	0.162	0.0001
1023 (slit edge)	0.0035	0.177	0.470	0.0028

At the settings each strategy actually runs with, the gap is the 50–500× seen throughout §7. Matched, it disappears — undistorted is even slightly better at slit centre, exactly what you'd expect from the cleaner baseline. So the difference was never about which pipeline's data is easier to fit. It's that native (and rectified) are allowed to nudge the assumed wavelength calibration by a small amount, and undistorted deliberately isn't.

What is that nudge actually correcting? A real, tiny, highly significant shift — a few thousandths of a spectral resolution element — that sits at essentially zero at the one along-slit position where keystone mixes the least, and grows toward both slit edges, exactly tracking how much true along-slit position keystone blends into a single row (§3). In other words: native's extra freedom is finding and quietly absorbing a small, real, keystone-linked registration error.

Undistorted is denied that same freedom on purpose. Handing it the same correction would make its chi-squared collapse to native's level too — but it would stop meaning anything as a zero-distortion reference, since it would be free to erase the very distortion signature it exists to isolate. A lower chi-squared for native, in other words, is a designed consequence of which pipeline gets a calibration escape valve — not proof its underlying fit, or its data, is somehow better.

Rectified gets the same escape valve as native and still fits worst of all three: its residual comes from real interpolation smearing (§4), a shape a smooth, low-order wavelength correction can't fix.

9 Open items

NOT STARTED

Multi-band retrievals

Every retrieval here is single-band. GeoCarb's real value often comes from combining bands — the O₂ A-band alongside a CO₂ band for path-length and aerosol correction, for instance — and nothing in this study yet tests whether that combination changes, or fixes, any of the bias shown in §7.

IN PROGRESS

Forward-model / calibration errors

Every result above gives the retrieval perfect knowledge of the real projection operator. A real instrument's geometric calibration is only known to finite precision, from noisy ground-test data, and can drift on orbit as the optical bench's temperature changes. Early exploratory work has begun on deliberately mismatching the retrieval's assumed geometry from the true one: a wavelength-calibration mismatch already shows a measurable, and partly self-correcting, effect, but a keystone/along-slit- position mismatch has no mechanism to enter these single-band retrievals at all yet — only the rectification step is structurally sensitive to it, and that extension isn't built.

Companion overview to the full methodology log — this page summarizes assumptions and results; the log carries the complete run-by-run record.